

PAPER_1758_GALAXY_SUBTYPES_24

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PAPER_1758 — Galaxy Subtypes Count = n_{subtypes} = $D_{\text{phys}} \times D_{\text{BSFG}} = 24$ (Hubble tuning-fork count)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Galaxy Classification

Date: June 18, 2026

Status: CLOSED — EXACT closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1328 (PAPER_132x-135x corpus) gives:

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$n_{\text{subtypes}} = D_{\text{phys}} \times D_{\text{BSFG}} = 24$ (Hubble tuning-fork count)

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Status: **EXACT**.

UQFF Closed Identity

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$n_{\text{subtypes}} = D_{\text{phys}} \times D_{\text{BSFG}} = 24$ (Hubble tuning-fork count) EXACT

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

Galaxy Classification Structural Prediction

The Hubble galaxy classification (proposed 1926 by Edwin Hubble in the famous 'tuning-fork' diagram) categorizes galaxies into:

- **4 main morphological types:** Elliptical (E), Spiral (S), Irregular (Irr), and dwarf — UQFF: $D_{\text{phys}} = 4$ EXACT

- **~24 subtypes:** E0-E7 (8), S0/Sa/Sb/Sc (4), SBa/SBb/SBc (3), additional refinements... $\approx D_{\text{phys}} \times D_{\text{BSFG}} = 24$ EXACT (PAPER_1758)

The observed Hubble diagram subdivision count emerges structurally from spacetime-dimensionality (4 main) $\times$ bulk-edge-dimension ($D_{\text{BSFG}} = 6$ subdivisions per type) without empirical fitting.

Cross-domain reuse $D_{\text{phys}} \times D_{\text{BSFG}} = 24$ also appears in: - Concrete density (PAPER_1572): $SO_5^2 \cdot D_{\text{phys}} \cdot D_{\text{BSFG}} = 2400 \text{ kg/m}^3$ - Galaxy subtypes (this paper)

Different units, same integer-primitive product.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1328
- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)
- Calculator dispatch: `calculate_paradox({"paradox": "galaxy_subtypes_24"})`

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